

SYDNEY TECHNICAL HIGH SCHOOL
(Est 1911)

MATHEMATICS EXTENSION II

HSC ASSESSMENT TASK 1

MARCH 2003

Time allowed : 70 minutes

Instructions :

- Show all necessary working in every question.
- Start each question on a new page.
- Attempt all questions.
- All questions are of equal value.
- Full marks may not be awarded for careless or badly arranged work.
- Approved calculators may be used.
- This test forms part of your HSC assessment.
- These questions are to be handed in with your answers.

Name : _____

Question 1	Question 2	Question 3	Total

Question 1

Marks

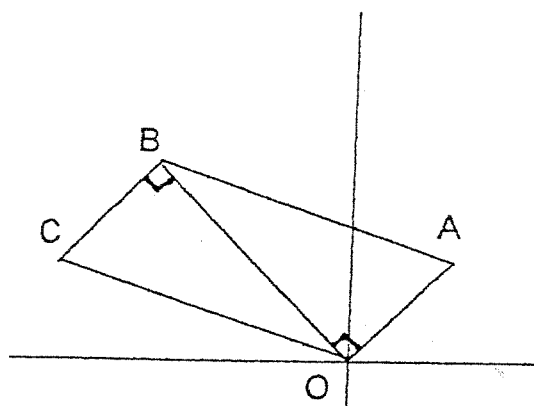
- a) Write $\frac{3+2i}{1+i}$ in the form $x + iy$ 2
- b) $(3+2i)(d+i)$ is real. Find the value of d 2
- c) In answering questions about the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ ($a > b$) Ting gave the following answers. 3
- i) eccentricity $0 \leq e \leq 1$
- ii) foci $(0, \pm ae)$
- Are Ting's answers correct. Give explanations for your choices.
- d) Find $|x + iy + 2|$ 2
- e) If z is the complex number $x + iy$ simplify $(z - \bar{z})^2$ 3
- f) i) Sketch the locus of z defined by $\arg(z+1) = \frac{\pi}{6}$ 2
- ii) Find z in the modulus argument form such that $|z|$ is a minimum 3

Question 2

Marks

- a) For the ellipse $4x^2 + 9y^2 = 36$ find
- i) eccentricity 2
 - ii) co-ordinates of foci 1
 - iii) equation of directrices 1
 - iv) sketch the ellipse marking all the necessary information 1

b)



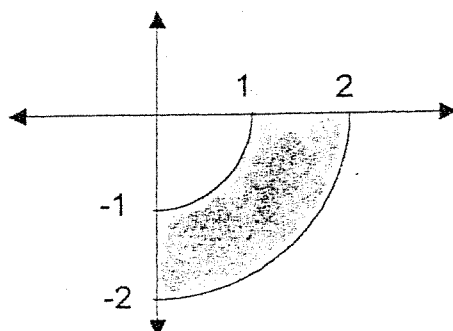
In the diagram $\angle CBO = \angle BOA = 90^\circ$ and $OB = 2OA$. If A is the complex number z

- i) Explain why B is the complex number $2iz$ 2
 - ii) If OABC is a parallelogram find the complex number for C 2
- c) Find the equation and sketch the locus of z if $\text{Im}(z^2) = 2$ 2
- d) Let A and B be the complex numbers z_1 and z_2 satisfying $|z_1| = |z_2|$
- i) Draw an Argand diagram showing the complex numbers z_1, z_2 and $z_1 + z_2$ (label C), z_1 and z_2 in the first quadrant. 2
 - ii) What type of figure is OACB 1
 - iii) Mark on your diagram the complex number $z_2 - z_1$ (label D) 1
 - iv) Use your diagram or otherwise to show that $\frac{z_1 + z_2}{z_2 - z_1}$ is imaginary 2

Question 3

Marks

- a) The complex number z is given by $z = -\sqrt{3} + i$. Find
- i) $\arg z$ 1
 - ii) $|z|$ 1
 - iii) z^7 in modulus argument form 2
- b) The roots of $z^6 - 1 = 0$ are $1, w, w^2, w^3, w^4, w^5$
- i) Find w (first complex root) in modulus argument form 1
 - ii) Plot all the roots on an Argand diagram 2
 - iii) By factorising or otherwise write down the equation whose roots are w, w^3 and w^5 1
- c) Give the inequalities which describe this region in the complex number plane. 3



- d) i) Prove that the equation of the tangent to the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ 3
- at the point $P(x_1, y_1)$ is $\frac{xx_1}{a^2} + \frac{yy_1}{b^2} = 1$
- ii) This ellipse meets the y axis at B and B^1 . The tangents at B and B^1 3
- to the ellipse meet the tangent at P at Q and Q^1 respectively.
- α) Draw a neat sketch labelling each of the points and showing the tangent.
- β) Prove $BQ \times B^1Q^1 = a^2$

Question 1

$$a) \frac{3+2i}{1+i} \times \frac{1-i}{1-i} = \frac{3-3i+2i+2}{2} = \frac{5-i}{2}$$

$$b) (3+2i)(d+i) = 3d+3i+2di-2$$

$$\text{If real } 3+2d = 0 \\ d = -3/2$$

c).

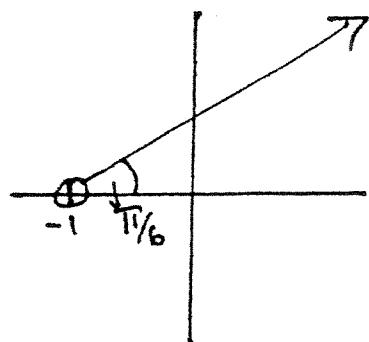
i) No $0 < e < 1$
 $e = 1$ is a parabola

ii) No $(0 \pm ae)$ are the co-ordinates of the focus of the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$

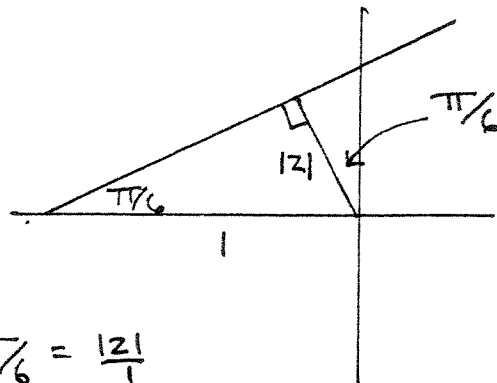
where $a < b$

$$d) |(x+2)+iy| = \sqrt{(x+2)^2 + y^2}$$

$$(z - \bar{z})^2 = [x+iy - (x-iy)]^2 \\ = [2iy]^2 \\ = -4y^2$$



ii)



$$\sin \pi/6 = \frac{|z|}{1}$$

$$\therefore |z| = 1/2$$

$$z = 1/2 \cos(\pi/2 + \pi/6) \\ = 1/2 \cos 2\pi/3$$

Question 2

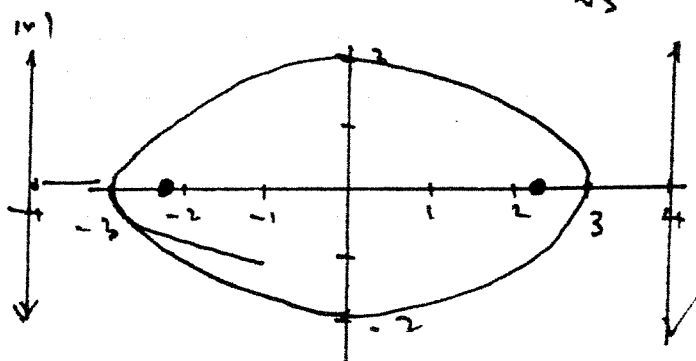
$$a) \frac{x^2}{9} + \frac{y^2}{4} = 1$$

$$ae = \sqrt{9-4} \\ = \sqrt{5}$$

$$i) e = \frac{\sqrt{5}}{3}$$

$$ii) \text{foci} = (\pm \sqrt{5}, 0)$$

$$iii) \text{directrices } x = \pm \frac{9}{\sqrt{5}}$$



b) i) $OB = 2OA$ and is rotated through 90° in an anticlockwise direction.

$$ii) \vec{Z} + \vec{AB} = 2i\vec{Z} \\ \vec{AB} = 2i\vec{Z} - \vec{Z}$$

$$OC \parallel AB \text{ and } OC = AB$$

$$\therefore \vec{OC} = 2i\vec{Z} - \vec{Z}$$

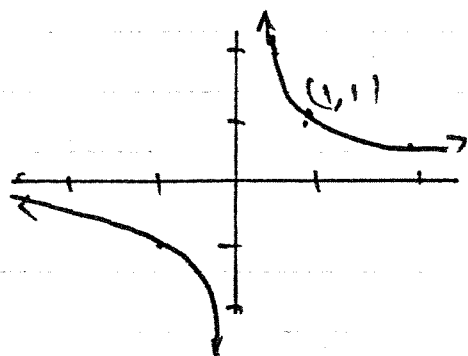
$\therefore C$ is the complex number $2iZ - Z$

c)

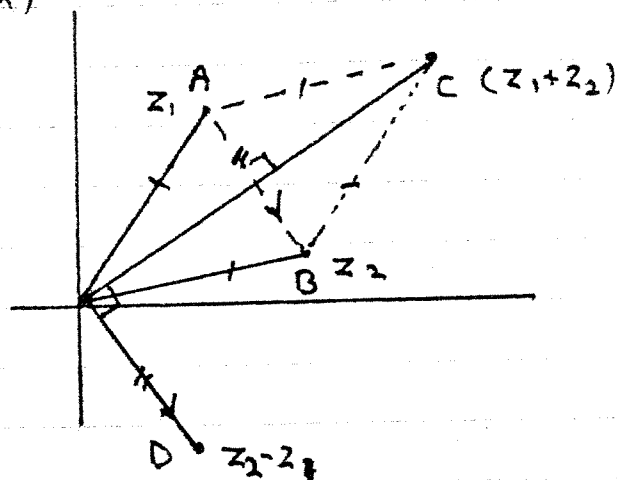
$$\begin{aligned} z^2 &= (x+iy)^2 \\ &= x^2 + 2xyi - y^2 \\ \therefore \operatorname{Im}(z^2) &= 2xy \end{aligned}$$

$$\therefore 2xy = 2$$

$$xy = 1$$



d)



i) rhombus

iv)

$$\text{Now } \arg(z_1+z_2) - \arg(z_2-z_1) = \pm \pi$$

$$\therefore \frac{z_1+z_2}{z_2-z_1} \text{ is imaginary}$$

Question 3

i) $\arg z = 5\pi/6$

ii) $|z| = 2$

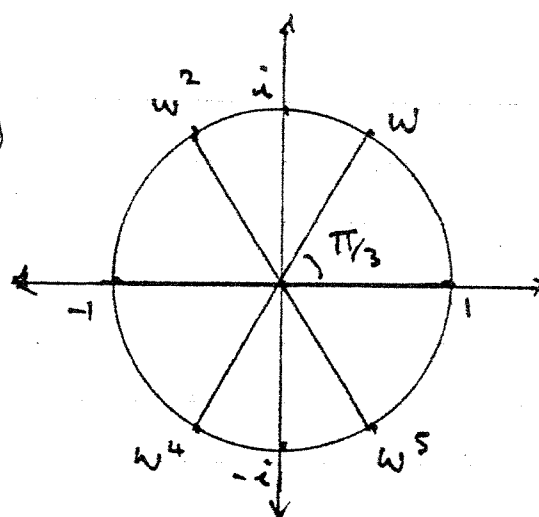
iii) $z' = (2 \cos 5\pi/6)^{11}$

$$= 128 \cos \frac{35\pi}{6}$$

$$= 128 \cos(-\pi/6)$$

b) i) $z = \cos \pi/3$

ii)



iii) $z^6 - 1 = (z^3 - 1)(z^3 + 1)$

w, w^3, w^5 are the roots of $z^3 + 1 = 0$

c) $1 \leq |z| \leq 2, -\pi/2 \leq \arg z \leq 0$

$$d) i) \frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$$

$$\frac{2x}{a^2} + 2y \frac{dy}{dx} = 0$$

$$\frac{dy}{dx} = -\frac{2x}{a^2} \times \frac{b^2}{2y}$$

$$= -\frac{b^2 x}{a^2 y}$$

$$\text{at } (x, y) = -\frac{b^2 x}{a^2 y}$$

Equation of tangent

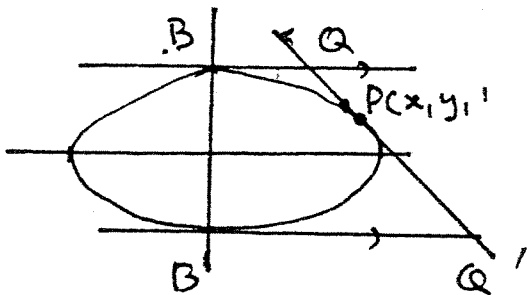
$$y - y_1 = -\frac{b^2 x_1}{a^2 y_1} (x - x_1)$$

$$\frac{yy_1 - y_1^2}{b^2} = -\frac{xx_1 + x_1^2}{a^2}$$

$$\frac{xx_1 + yy_1}{a^2} = \frac{x_1^2 + y_1^2}{b^2}$$

but (x, y) lies on ellipse

$$\therefore \frac{xx_1}{a^2} + \frac{yy_1}{b^2} = 1$$



Equation of tangent at B

$$y = b$$

$$\text{at } Q \quad \frac{xx_1}{a^2} + \frac{yy_1}{b^2} = 1$$

$$x = \left(1 - \frac{by_1}{b^2}\right) \frac{a^2}{x_1}$$

$$= \frac{a^2 b^2 - a^2 b y_1}{b^2 x_1}$$

Likewise at Q'

$$x = \frac{a^2 b^2 + a^2 b y_1}{b^2 x_1}$$

Now $BQ \times B'Q'$

$$= \left(\frac{a^2 b^2 - a^2 b y_1}{b^2 x_1}\right) \left(\frac{a^2 b^2 + a^2 b y_1}{b^2 x_1}\right)$$

$$= \frac{a^4 b^4 - a^4 b^2 y_1^2}{b^4 x_1^2}$$

$$= \frac{a^2 (a^2 b^2 - a^2 y_1^2)}{b^2 x_1^2}$$

Now since (x, y) lies on the curve

$$\frac{x_1^2}{a^2} + \frac{y_1^2}{b^2} = 1$$

$$\therefore b^2 x_1^2 + a^2 y_1^2 = a^2 b^2$$

$$\therefore \frac{a^2 (a^2 b^2 - a^2 y_1^2)}{a^2 b^2 - a^2 y_1^2}$$

$$= a^2$$

$\therefore$ Proven